

A modular causal adaptation architecture for machine learning on data-scarce rare disease

Problem. Rare diseases are called orphan diseases because each affects few patients, leading to fewer clinical trials, making them data-scarce to model and leaving pharmaceutical companies a smaller market to capture. Individually negligible, they are not so collectively. Perturbation foundation models condition on experimental context via a learned lookup table, so a context not encountered during pretraining has no embedding and no prediction, and a patient-derived line with a particular lesion will never be screened at that scale. Starting from scratch is not an option. We therefore ask a more restricted, verifiable question: where in the network does the disease enter, from a few observations in an unseen context?

Approach. Learning rare-disease biology from scratch is inefficient. Instead, we isolate specific pathological shifts within a pre-wired model of general biology using a modular loop where each component executes a distinct task:

- **Frozen Causal Foundation:** a pre-trained structural causal model defining baseline cell wiring W_0 and per-context gains a_c over an immutable, supplied topological sorting.
- **Non-Linear System Dynamics:** cellular state progression modelled using a generic saturating nonlinearity, where $x_i = \tanh\left(\sum_j W_{ji} a_{c,i} x_j + b_i\right) + \varepsilon_i$.
- **Rank-One Causal Adapter:** the central diagnostic modifier, parameterizing the pathological mutation as a low-rank deviation matrix, where $W_{\text{lesion}} = W_0 \odot a_c + uv^\top$.
- **Structural Lesion Allocation:** a soft one-hot routing vector over the network’s entry node, where $u = \text{softmax}(u_{\text{logits}}/\tau)$ with an annealed temperature parameter τ .
- **Sparse Network Rewiring:** a downstream perturbation impact vector v constrained by an ℓ_1 -norm penalty ($\|v\|_1$) to prevent dense parameter leakage.
- **Parametric Dimension Reduction:** factorization reduces the parameter count from G^2 to $2G$ continuous parameters, permitting sample-efficient model estimation with only $k \sim 10$ unique observations.
- **Counterfactual Engine:** an interventional oracle scoring candidate downstream phenotypic outcomes by executing automated $do(\cdot)$ mutations across the adapted structural graph.
- **Epistemic Uncertainty Tracking:** a six-member bootstrap ensemble recording variations in localized parameters to flag ungrounded, low-confidence graph predictions.
- **Active Experiment Selection:** an acquisition loop optimizing sample selection by scheduling lab experiments for target nodes exhibiting the highest ensemble disagreement regarding downstream expression profiles.

Results. On simulated networks with $G=50$, lesion-node localization improves substantially as the number of observations k increases, rising from 0.00 accuracy at $k=5$ to 0.72 at $k=40$, compared with a chance level of 0.02. Under the same regularization, an unconstrained sparse ΔW achieves only 0.11 accuracy, suggesting that the improvement is driven by the rank-one structural prior rather than sparsity alone. However, the method does not generalize to the real Sachs dataset: when one reagent regime is held out and its known target is hidden, lesion recovery remains at chance. Experiments injecting known lesions into networks of increasing size indicate that network size and lesion identifiability are major limiting factors: lift over chance increases from 1.22 at $G=11$ to 6.48 at $G=50$, while at $G=11$ a rank-one deviation is barely distinguishable from perturbations of existing network weights. Corrupting the supplied causal ordering produces only a modest decline in accuracy but substantially reduces reliability, while structural metrics fail to provide an early warning of this degradation. Finally, ranking interventions by predicted rescue performs no better than an unadapted baseline, suggesting that lesion localization and therapeutic ranking are separable tasks, with the current adapter benefiting the former but not the latter.

Context. The proposed framework rests on a singular hypothesis: a rare disease is a localized, low-rank deviation from a global biological network the model already understands. While prior work attempts to place unseen contexts by encoding static, unperturbed profiles, we estimate the structural deviation directly, enabling sample efficiency down to ten observations. The evaluations reveal that structural identifiability scales directly with network size and favours incoming over outgoing lesions, but degrades sharply under topological order misspecification. Further, this structural dependency explains why root localization succeeds where downstream intervention ranking fails, as therapeutic sequencing is primarily dictated by the shared baseline topology rather than localized path corrections. Current limitations include the requirement of a pre-specified acyclic topological sorting, the omission of feedback loops, and a validation pipeline that closes the active learning loop via a digital simulator rather than direct wet-lab execution.

References

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